

DES303 Week 1: Inspiration & Ideation.

8 min read · Mar 10, 2025



Livi Lee

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March 6th, 2025: My first day of DES303.

Coming into DES303, I had some important goals. From conversations with tutors and lecturers, as well as previous Capstone students, I had heard a lot about the difficulties that many final-year students faced when it came time to make their Capstone project. This ‘experiment-based’ course meant that we had some extra time to develop the skills to pull off a more advanced Capstone project.

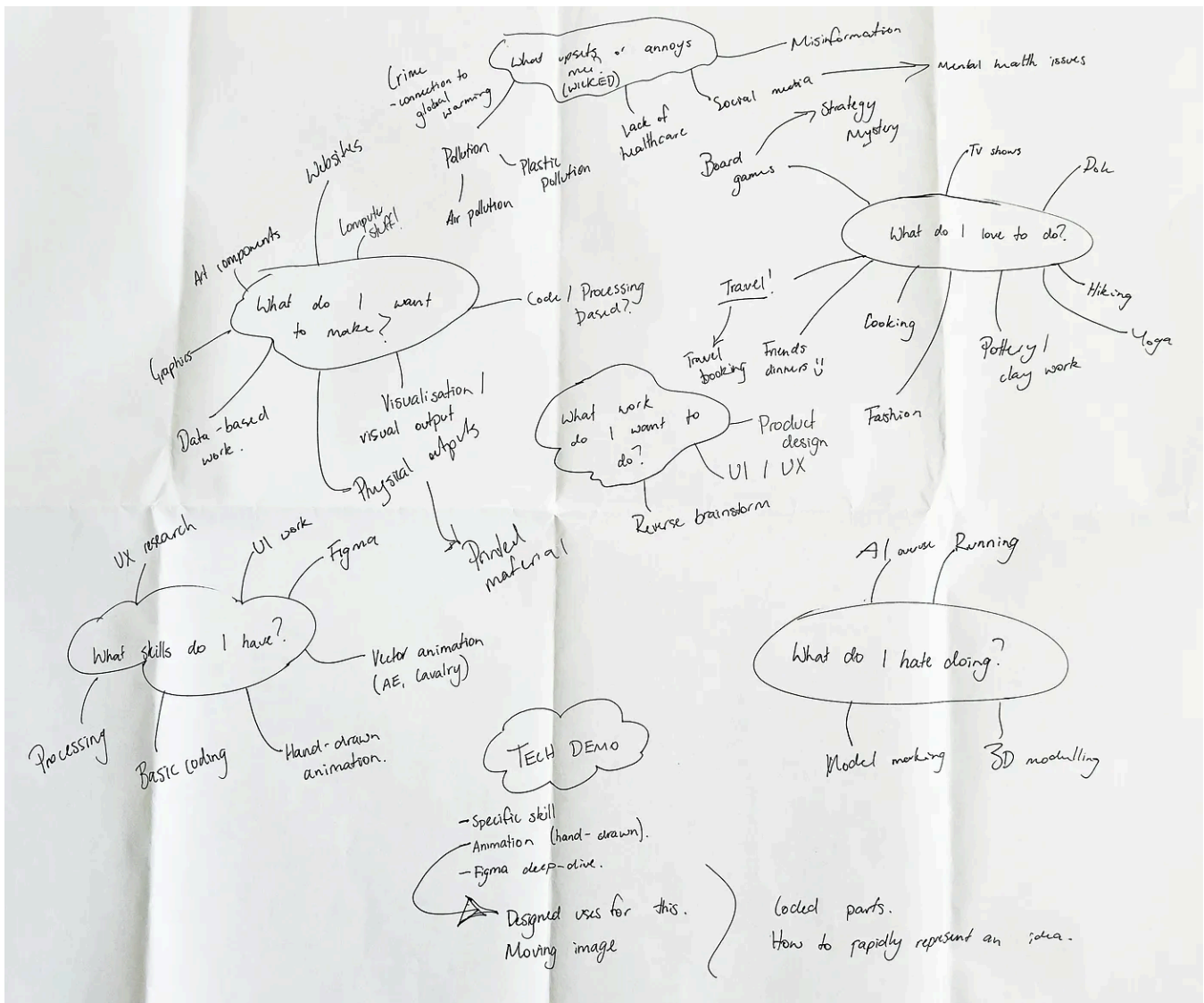
My philosophy here: if, as I write this, I have all the knowledge and skills needed for the technical execution of my final project, I’ve failed to sufficiently push myself. It’s an unusual (but exciting) feeling; *hoping* that I don’t yet have the technical skills to achieve what I want to do! I’m looking forward to having some time to experiment with different mediums before settling on a final output.



Me and Roma at the 2024 ReDesign Showcase.

Having worked extensively on last year's ReDesign Showcase, where the Capstone projects of 2024 graduates were displayed, I feel uniquely prepared to work on this project. I'm particularly aware that I will need to produce a large-scale poster of my final output, alongside having a small booth to showcase my work. From my experience, graduates who produced high-fidelity physical outputs did much better at Showcase than those who produced exclusively digital outputs. Thus, it's important that I keep in mind those final physical outputs, and what the experience of this project will be like for viewers at ReDesign.

Part I: Beginning Ideation



The mind map I made during DES303 studio time.

Shown above is the mind map I made during the studio time in DES303. This mind map was useful for getting some of the ideas on paper that I've had floating around in my head over the summer. However, I feel like I only scratched the surface in this mind map, and I haven't yet come up with any specific experiment ideas. I'll keep adding to this over the next week as I continue ideating.

Key takeaways from my preliminary mind map:

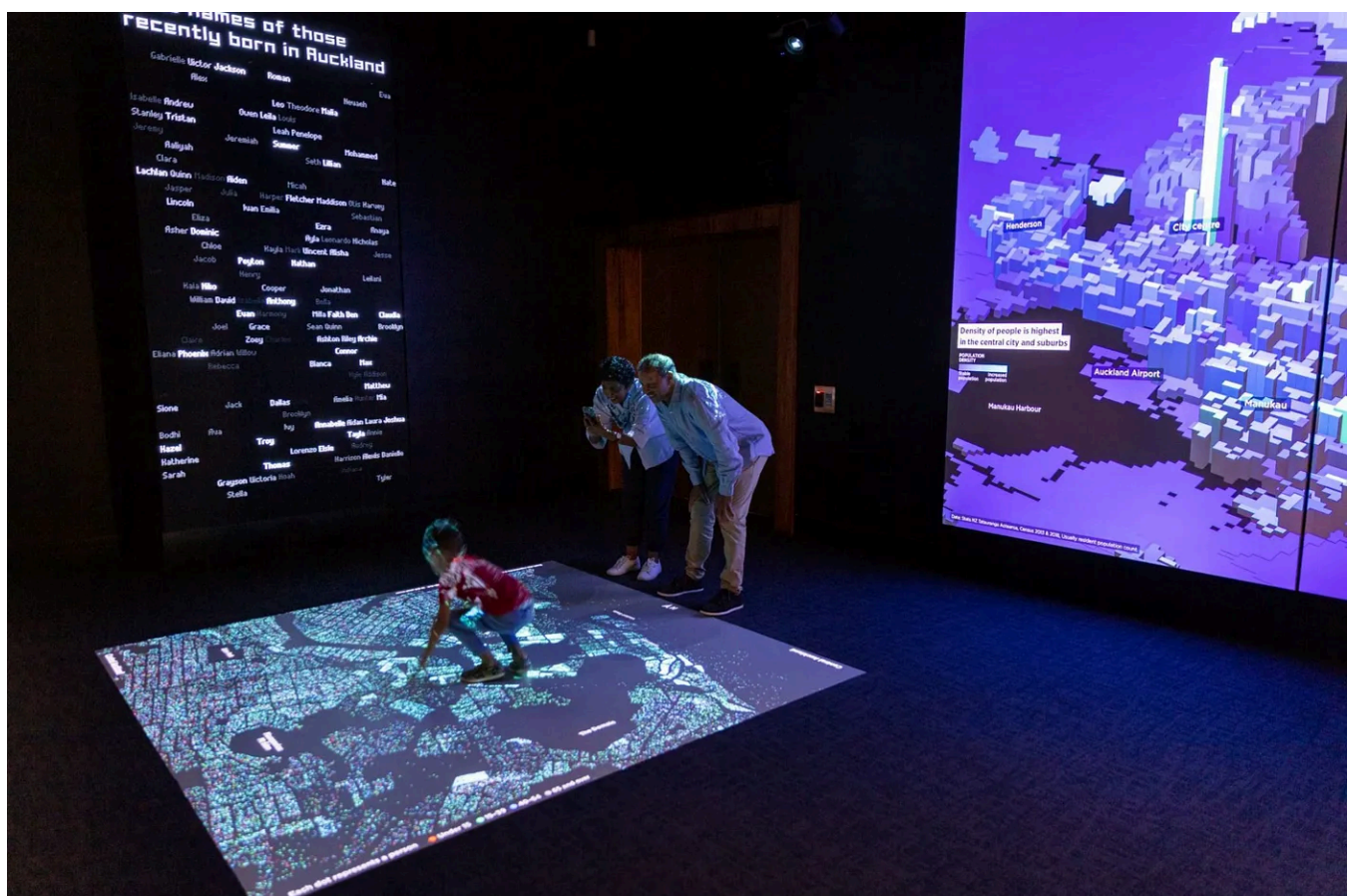
- I want to use digital tools to create my experiments.
- I want the output of these experiments to be primarily interactive visual work. For instance: small games and interactive displays. I am also interested in experimental apps and websites, as well as vector-based animation.
- I want to be able to create physical artifacts from my digital visualisations. For instance: posters, cards, booklets. I found that students with physical outputs

did much better at the exhibition last year than their peers with only digital outputs.

- I have three main 'WICKED problems' of interest: local environmental pollution, the sociological effects of climate change, and digital misinformation on social media.

Part II: Inspirations

An idea that I'm particularly keen to explore is an interactive data-based display. This idea was inspired by one of my lecturers in Communications, Ethan Plaut, who worked on the Data Room in Auckland War Memorial Museum (where I also worked as a Visitor Host) last year. This exhibit incorporates different types of data from around Auckland and projects data visualisations on the walls, ceiling, and floor. It's highly engaging, and working at the museum, I saw how much people loved — and learnt from — this exhibit.



[Stories of Auckland Tāmaki Herenga Waka is open at Auckland Museum | Auckland for Kids](#)

This inspired me to look more into data visualisation and data-based exhibition work. Although the Data Room was fascinating, I felt that it lacked a 'purpose'. The data that was being collected — about public transport routes, wastewater pipes,

Part III: Challenges

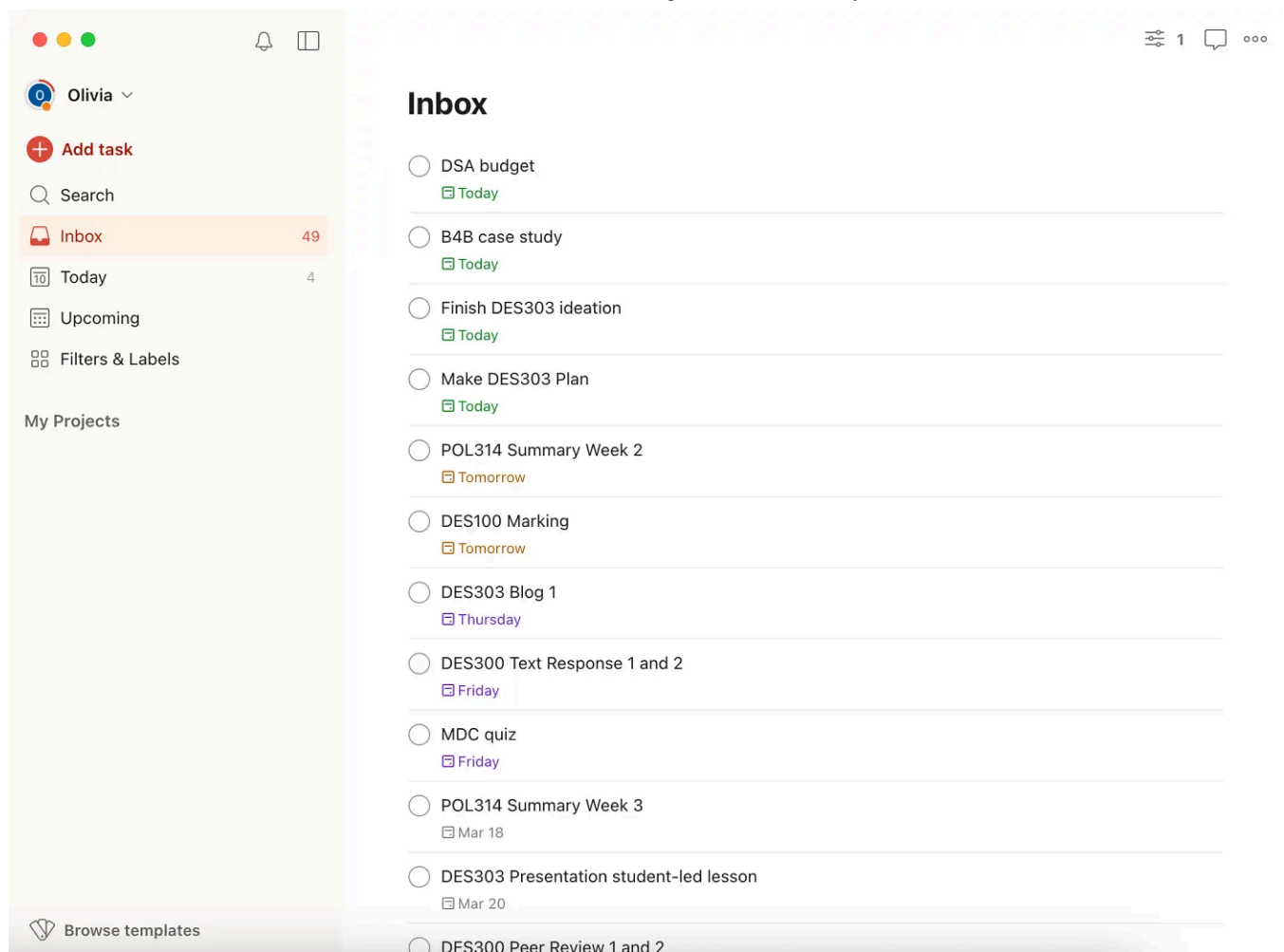
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Part IV: Tech Demo Ideation

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I started out my tech demo ideation with three main ideas. These are skills that I have developed at university and as part of my work over the last few years.

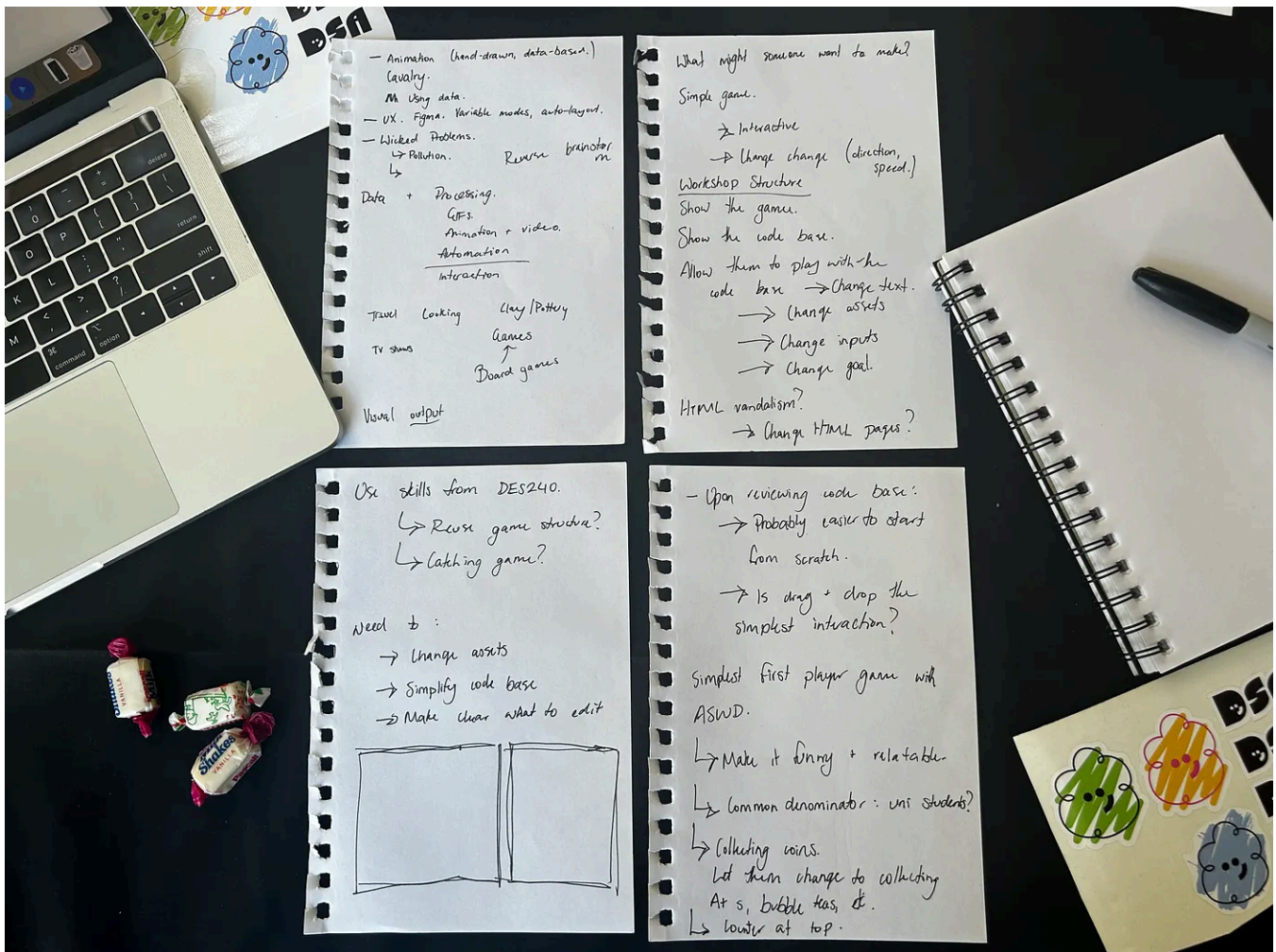
- 1. Advanced Figma techniques.** I worked extensively using Figma during my internship over the summer. As it's an industry standard programme, learning how to use more advanced features (eg. variable modes, design systems) could be very useful to my classmates. However, it is likely that my peers will have

varying existing abilities in Figma. I don't want to teach a demo that is too hard or too easy for my peers.

2. **Interactive game making in Processing.** I really enjoyed learning about Processing during DES240 last year; however, many of my classmates found it really daunting to even begin learning how to use code to create visualisations. By making small, adaptable games, I could demonstrate some basic programming techniques. My level of proficiency in Processing is probably lower than either of my other options, but this could be a benefit; I could expand my learning at the same time as I am teaching others.
3. **Hand-drawn animation.** This is something that I spent the summer practicing. One benefit of hand-drawn animation is that it can be used to communicate so many different ideas; if you can draw it, you can animate it! This could make it very useful for sharing with my peers. However, I'm not sure how much I could 'teach' in a tech demo — the techniques that I use are pretty simple, it just comes down to practice.

After considering these options, I think that interactive game making in Processing is the best choice for my tech demo. There are several reasons I've chosen this medium:

- Processing is made to be simple for beginners, so I can easily teach it during a 15-minute demo.
- When I've talked to my peers before, many of them found coding daunting or had never tried it before. If I can make it more accessible, it could open doors that they haven't considered.
- Learning basic programming is easy but knowledge-based; you have to know *what* to learn to be effective (as opposed to practice-based skills where the key is just time!). I think my demo will be most effective for Processing, as opposed to me teaching the other skills.
- Visualised coding closely aligns with what I am interested in creating for my technical experiments. By practicing my coding for the tech demo, I will also be upskilling for future experiments.

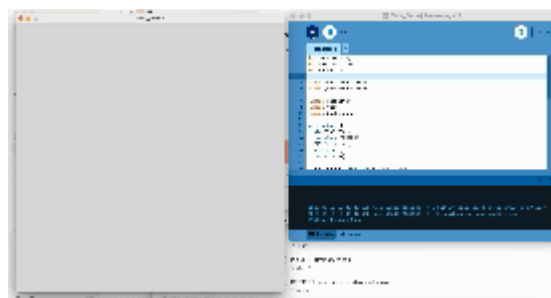


My notes brainstorming what to teach my peers about Processing, and how to do so.

To test my idea for the tech demo, I have started building a basic interactive game in Processing. My goals here are to:

- Make a game with the simplest code possible.
- Make a highly adaptable game so that my peers can experiment and play with the code to make it their own.

With very little code and simple commands, I was able to make this random image appear with a water background. Students would be able to change the background, image, speed, keys, and collectable items. They could also change the game's goal through code.



A GIF showing the first lo-fi test of what my tech demo output could be.

The goal of my tech workshop would be to show my peers how easy it is to get started with Processing, and teach them some basic commands to make their own games.



```
1 int Counter = 0;
2 int numItems = 10;
3 int speed = 10;
4
5 float xPosition = 500;
6 float yPosition = 500;
7
8 PImage character;
9 PImage item;
10 PImage background;
11
12 void setup(){
13   size(700,700);
14   imageMode(CENTER);
15   fill(0, 0, 0);
16   noStroke();
17   textSize(30);
18
19   background = loadImage("background-1");
20   character = loadImage("character-sprite-1.jpg");
21   item = loadImage("item-1.png");
22
23 }
24
25 void draw(){
26   imageMode(CORNER);
27   image(background, 0, 0, 700, 700);
28   imageMode(CENTER);
29   image(character, xPosition, yPosition);
30 }
```

2025-03-10 15:00:23.684 java[59911:2659811] +[IMKClient subclass]: chose IMKClient_Modern
2025-03-10 15:00:23.684 java[59911:2659811] +[IMKInputSession subclass]: chose

I was able to make these two images appear and move with very basic code.

Over the next two weeks, I will continue developing this idea and testing what can be achieved with relatively simple code.

Note: I discovered that we know which students we will be presenting this tech demo to (ie. they are not randomly assigned on the day.) Thus, I will check in with these four peers to understand more about what their coding proficiency is to see what difficulty level I should use.

Part V: Experiment Ideation

As explained above, a theme that I want to explore in my coursework this semester is interactive coded projects. I have some experience creating websites using HTML and CSS, as well as doing some projects in Processing last year. My internship over the summer also gave me experience working with developers and understanding what is and what is not possible with code. This gives me a solid foundation on which to continue building my skillset.

One thing that I really like about coded designs are how scalable they are. When set up correctly, one tiny change in the code can create a big impact in the final output. To me, this means that it's a more effective tool for visual communication.

From my initial experiment ideation session, there's some ideas that I immediately want to follow up. These include:

- Processing-based interactive games. This is something that I am pursuing as part of my tech demo. If successful, I hope to extend it in the future.
- Using physical sensors to create data visualisations. The Design Lab has several sensors and trackers that could be interesting here: moisture and humidity sensors, temperature sensors, proximity sensors. If I can combine these with Processing-based visualisations, I could create some really interesting work.
- Using data visualisations to create graphic artifacts (posters, playing cards, etc). I know I want to create physical artifacts as part of my DES304 capstone, so it makes sense to begin experimenting with printed formats now.

I will continue ideating about possible experiment ideas, and refining the above ideas into a solid plan, in my blog post next week. Mā te wā!

[Des303](#)[Design](#)[Reflection](#)

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Written by Livi Lee

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Kia ora, I'm a designer at the University of Auckland dedicated to transforming innovative ideas into beautiful, functional visual realities.

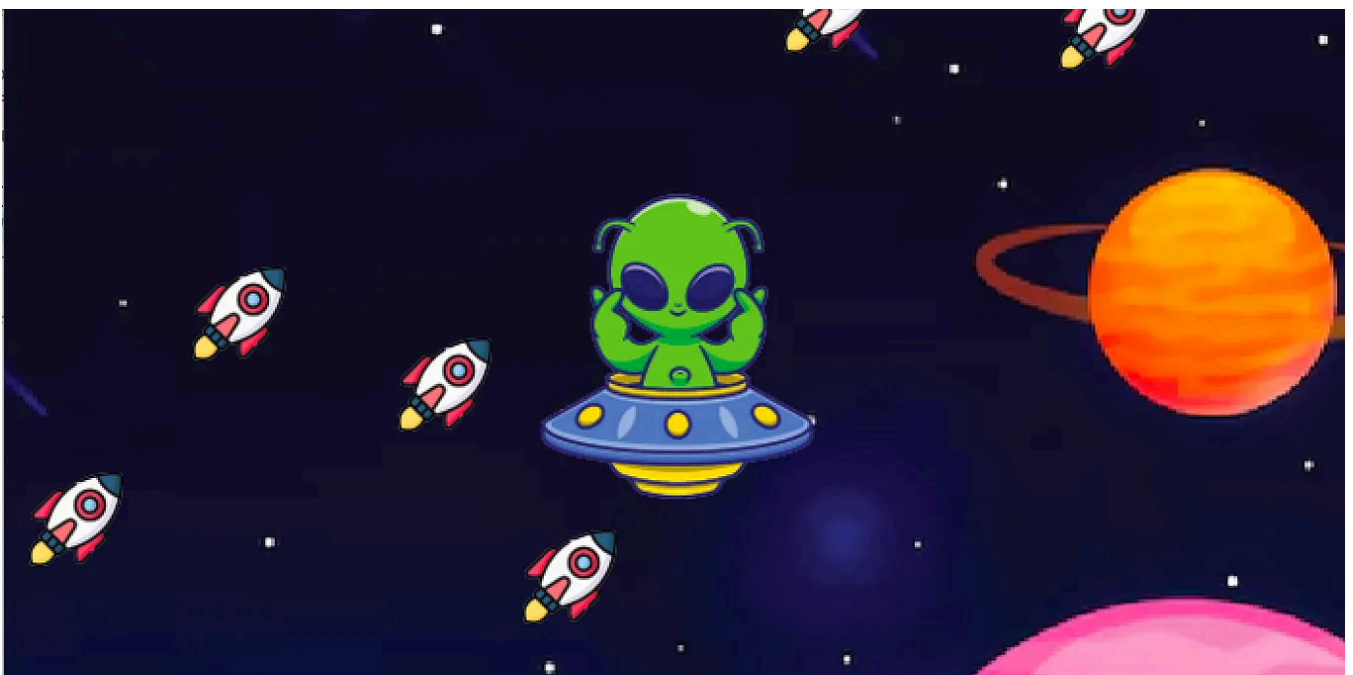
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What are your thoughts?

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Livi Lee

DES303 Week 2: Tech Demo Prep & Experiment Ideation

I needed to hit the ground running this week. With our tech demos just over a week away, experiment ideation to do, and blog posts to write...

Mar 17, 2025



DES303 – Design Research Practice

March 2025



Reviewing Your Design Research Journey

Make a chronology of your experiments

- Extract your action plans from each blog post (summarise if needed)
- Evaluate your progress with each plan
- Select screenshots/images/visuals from each experiment
- Use these elements to build a more concise narrative of your journey
- Describe your final experiment and the key factors leading to it

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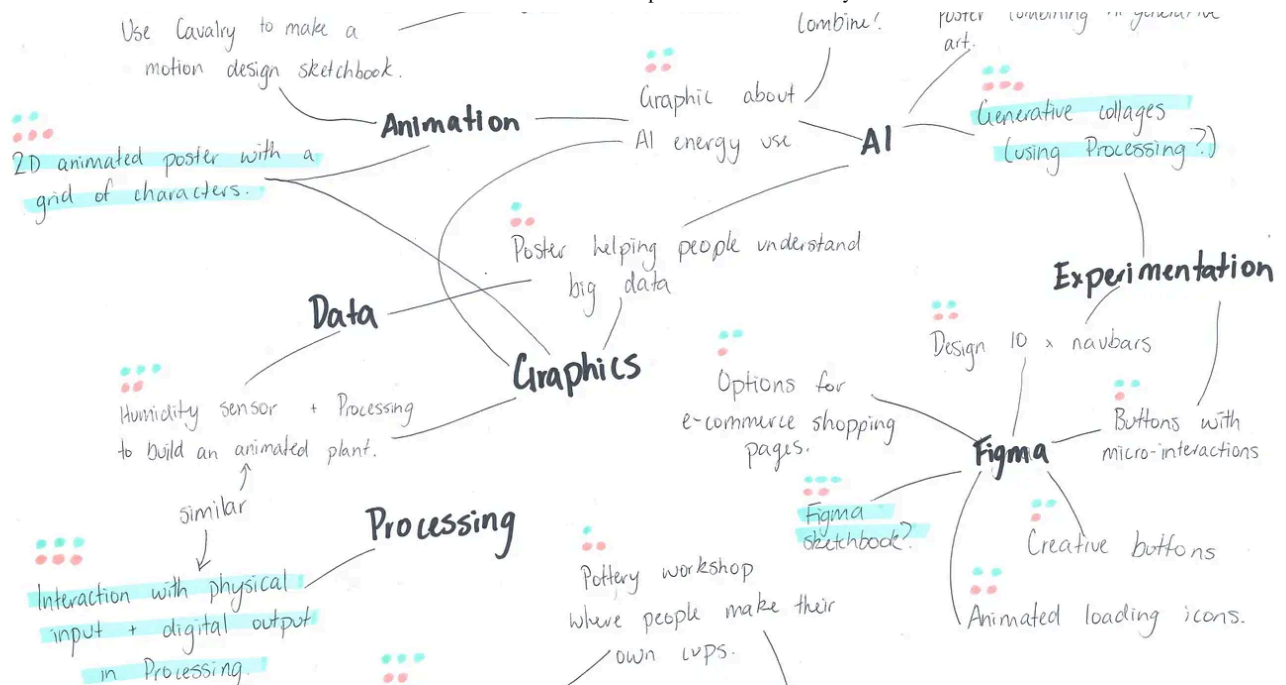
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DES303 Week 10: Website Changes & Blog Review

Welcome to my final blog for DES303! This week I'll be wrapping up my experimentation and reviewing my prior blogs to help plan my final...

May 27, 2025



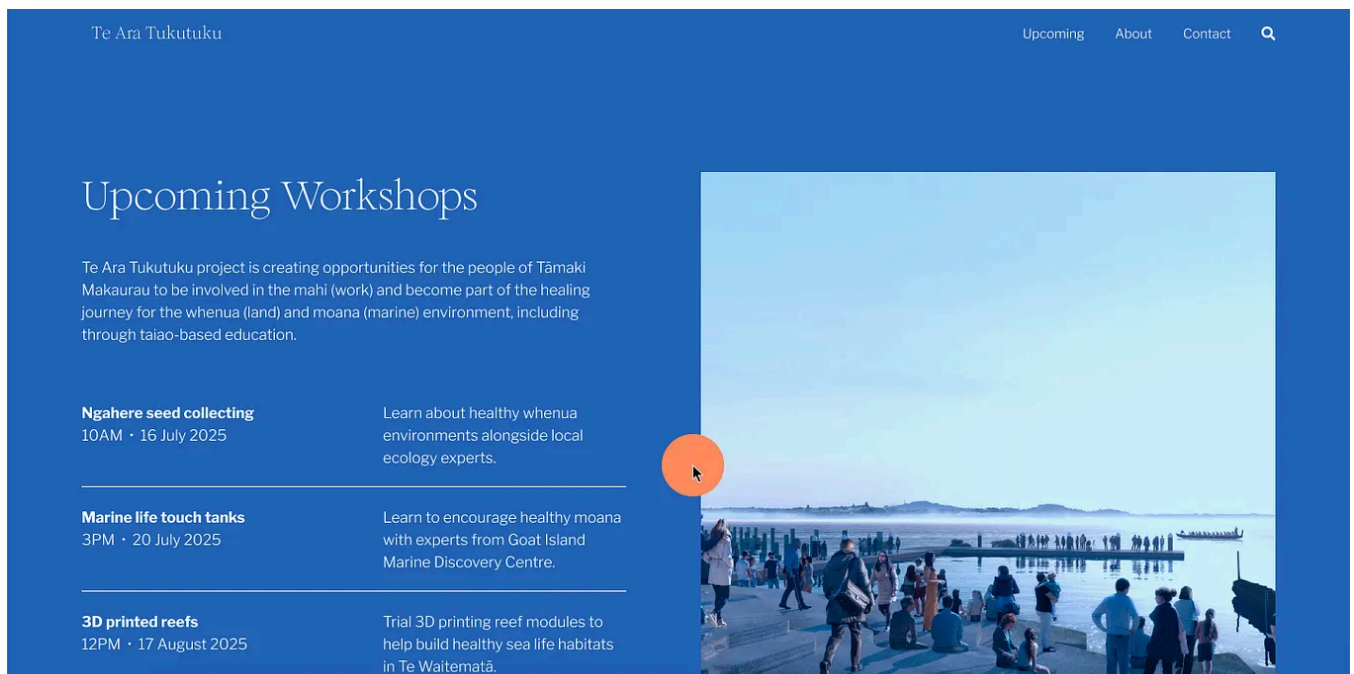


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DES303 Week 3: Tech Demo Reflection and Experiment Planning

With the tech demo behind us, this week was a little more relaxed. I got to invest more time into my experiment ideation and planning. I'm...

Mar 23, 2025



Livi Lee

DES303 Week 9: Final Pivot & Final Crit!

As I discussed in my blog last week, I've started a new job that has significantly reduced the amount of time I can spend in the Design Lab...

May 24, 2025

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:2510.01171v3 [cs.CL] 10 Oct 2025

ABSTRACT

Post-training alignment often reduces LLM diversity, leading to a phenomenon known as *mode collapse*. Unlike prior work that attributes this effect to algorithmic limitations, we identify a fundamental, pervasive data-level driver: *typicality bias* in preference data, whereby annotators systematically favor familiar text as a result of well-established findings in cognitive psychology. We formalize this bias theoretically, verify it on preference datasets empirically, and show that it plays a central role in mode collapse. Motivated by this analysis, we introduce **Verbalized Sampling (VS)**, a simple, training-free prompting strategy to circumvent mode collapse. VS prompts the model to verbalize a probability distribution over a set of responses (e.g., “Generate 5 jokes about coffee and their corresponding probabilities”). Comprehensive experiments show that VS significantly improves performance across creative writing (poems, stories, jokes), dialogue simulation, open-ended QA, and synthetic data generation, without sacrificing factual accuracy and safety. For instance, in creative writing, VS increases diversity by $1.6\text{--}2.1\times$ over direct prompting. We further observe an emergent trend that more capable models benefit more from VS. In sum, our work provides a new data-centric perspective on mode collapse and a practical inference-time remedy that helps unlock pre-trained generative diversity.

Problem: Typicality Bias Causes Mode Collapse

Tell me a joke about coffee.

Solution: Verbalized Sampling (VS) Mitigates Mode Collapse

Different prompts collapse to different modes:

In Generative AI by Adham Khaled

Stanford Just Killed Prompt Engineering With 8 Words (And I Can't Believe It Worked)

ChatGPT keeps giving you the same boring response? This new technique unlocks 2× more creativity from ANY AI model—no training required...

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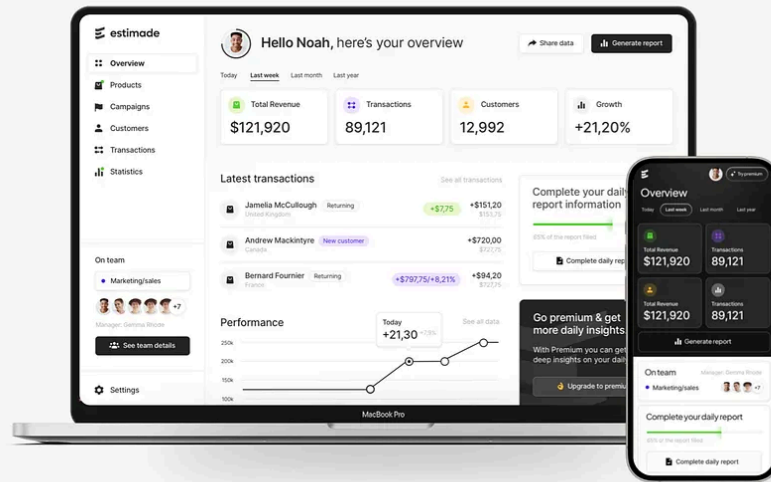
 In Artificial Corner by The PyCoach

The Best AI Tools for 2026

If you're going to learn a new AI tool, make sure it's one of these

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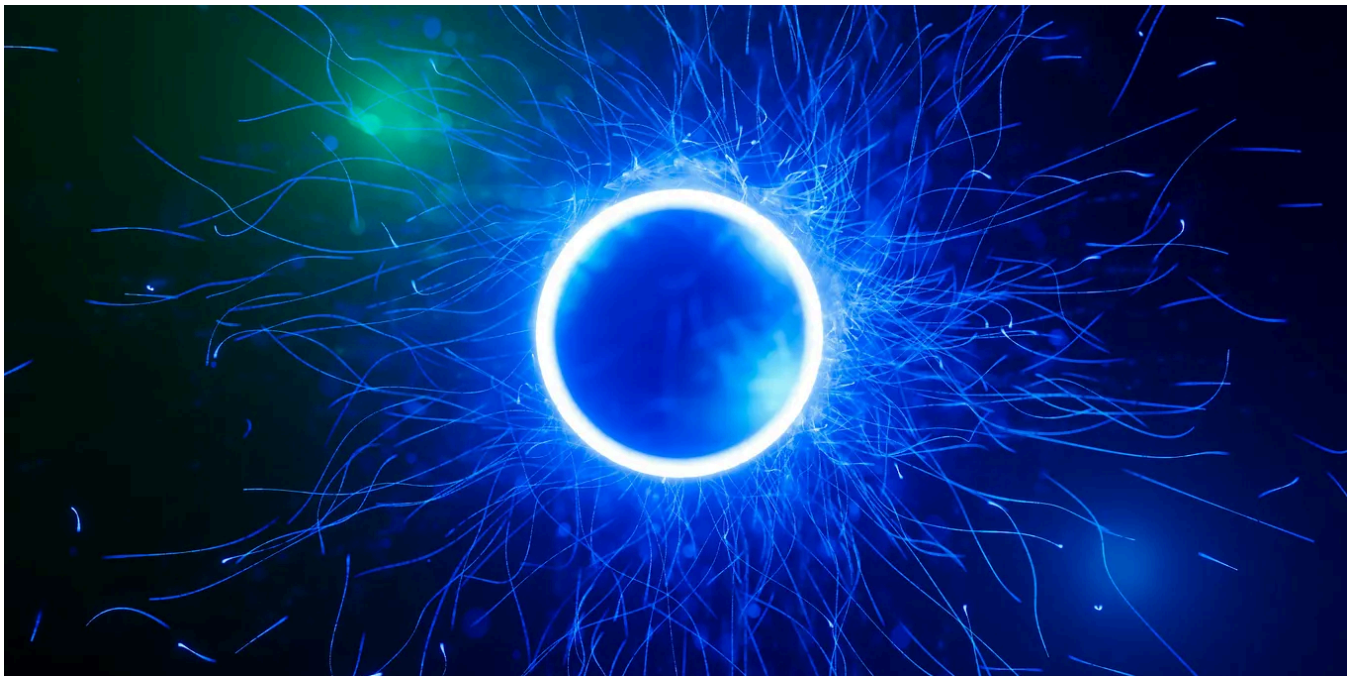
 Michal Malewicz 

The End of Dashboards and Design Systems

Design is becoming quietly human again.

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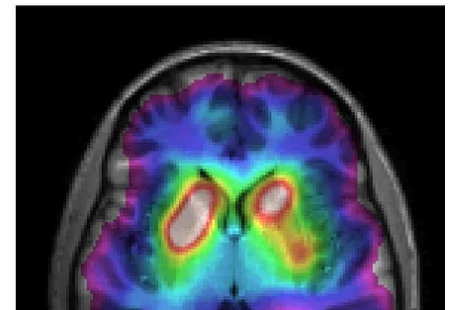
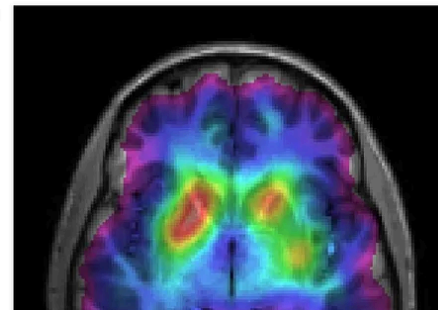
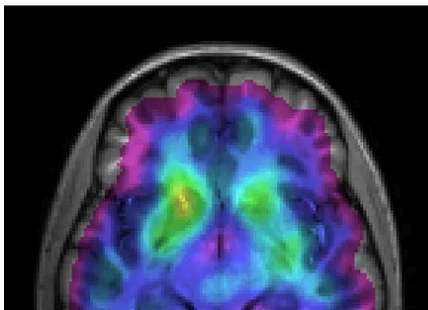
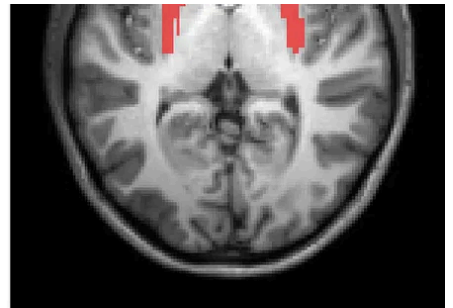
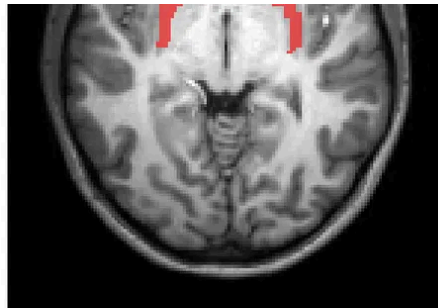
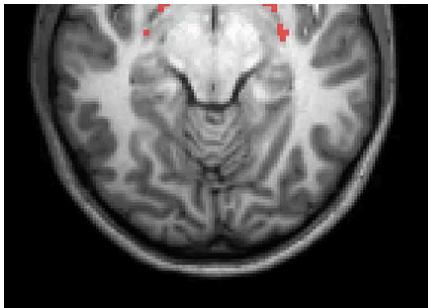


Will Lockett

The AI Bubble Is About To Burst, But The Next Bubble Is Already Growing

Techbros are preparing their latest bandwagon.

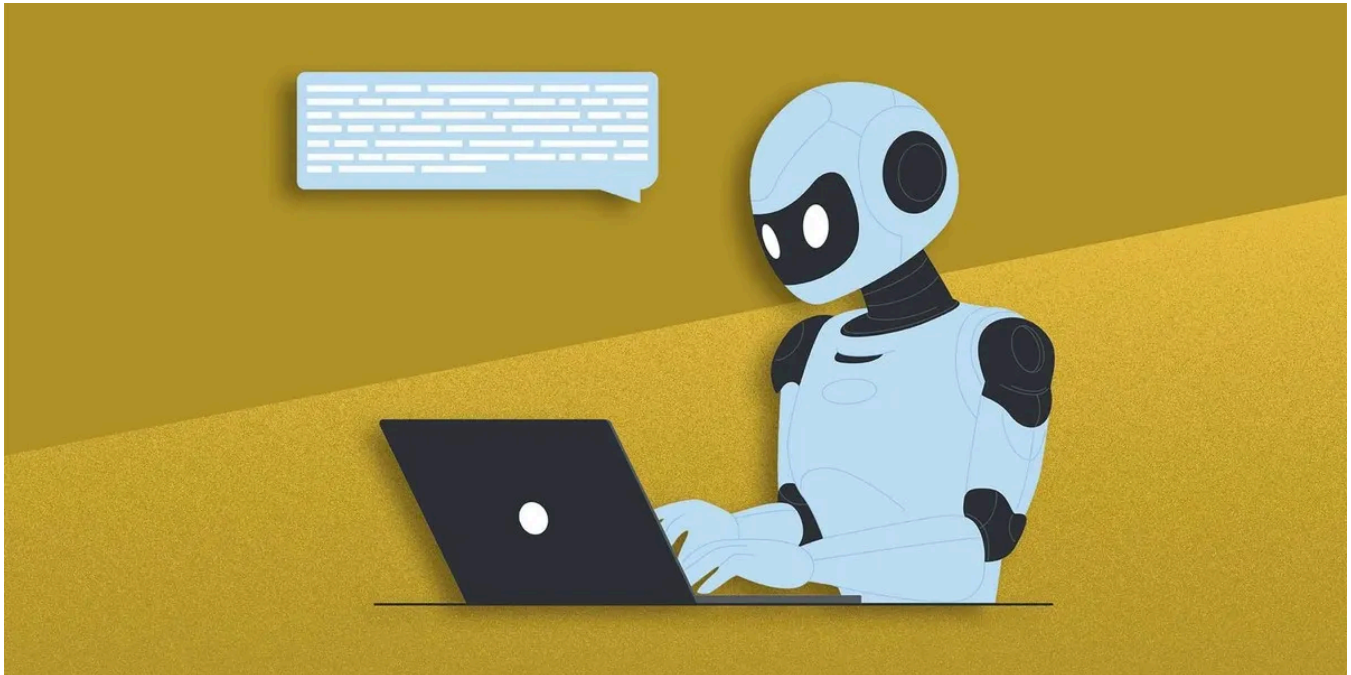
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